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BIDIRECTIONAL NEURAL MACHINE TRANSLATION BETWEEN STANDARD BANGLA AND THE CHATGAYA DIALECT

USING FINE-TUNED mBART-50

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1.INTRODUCTION

- **Chatgaya** is a major Bangla dialect spoken in the Chattogram region, yet it remains a low-resource language with limited parallel corpora and machine translation resources.
- **This research develops a bidirectional neural machine translation system** between Standard Bangla and Chatgaya using fine-tuned mBART-50 models, supported by a manually curated parallel corpus, lexicon-based data augmentation, and native speaker validation.

2.RESEARCH MOTIVATION

Recent advances in Neural Machine Translation (NMT) have significantly improved translation quality for many high-resource languages. However, regional languages and dialects such as the Chatgaya dialect continue to receive limited attention due to the scarcity of high-quality parallel corpora and linguistic resources. This research aims to address these challenges by developing a bidirectional neural machine translation system between Standard Bangla and the Chatgaya dialect.

3. RESEARCH OBJECTIVES

- ✓ Construct a manually curated parallel corpus for Standard Bangla and the Chatgaya dialect.
- ✓ Develop separate translation models for Standard Bangla→Chatgaya and Chatgaya→Standard Bangla.
- ✓ Investigate the effectiveness of canonical and augmented datasets for low-resource dialect translation.
- ✓ Evaluate translation quality using automatic evaluation metrics and human evaluation.
- ✓ Develop a Flutter-based prototype translation application integrated with a FastAPI backend for real-time bidirectional translation.

4. PROBLEM STATEMENT

Existing multilingual translation systems for the Chatgaya dialect are limited by the lack of publicly available parallel corpora, benchmark datasets, and comprehensive experimental studies. Moreover, the linguistic differences in vocabulary, pronunciation, and sentence structure make accurate bidirectional translation between Standard Bangla and Chatgaya particularly challenging.

5.RESEARCH QUESTIONS

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"Building AI for language diversity and inclusion"

- Manually Curated Data
- Native Speaker Validated
- Linguistically Informed Augmentation
- Real-world Deployment

DATASET CONSTRUCTION PIPELINE

1 Canonical Corpus Construction
- Manually create parallel sentence pairs.
- Standard Bangla ↔ Chatgaya Dialect.

2 Lexicon Development
- Build lexicon of core words.
- Include multiple Chatgaya variations.
- POS and meaning mapping.

3 Native Speaker Validation
- Survey native speakers.
- Validate sentence correctness.
- Collect real usage variations.

4 Lexicon-based Augmentation
- Replace words using lexicon variations.
- Generate multiple diverse sentences.
- Maintain semantic consistency

5 Final Parallel Corpus
- Remove duplicates.
- Quality filtering.
- Final high-quality parallel dataset.

6 Dataset Ready
- Clean, balanced and high-quality corpus.
- Ready for model training and evaluation.

CANONICAL CORPUS
2,020 Sentence Pairs (Manual)

LEXICON
~50 Core Words with Variations

FINAL DATASET (AFTER AUGMENTATION)
BN → CTG : 2,507 pairs
CTG → BN : 2,507 pairs
Total : 5,014 pairs

DATA QUALITY
- Balanced
- Validated
- Diverse
- High-quality

Balanced Domains 10 semantic categories covering daily life situations.

Native Validation Ensures authenticity and real-world usage.

Lexicon Augmentation Generates diverse yet meaning-preserving variations.

High Quality Corpus Ready for fine-tuning mBART-50 and robust evaluation.

OVERALL RESEARCH WORKFLOW

1 Data Collection & Preparation
Collect sources, define domains, and prepare raw data.

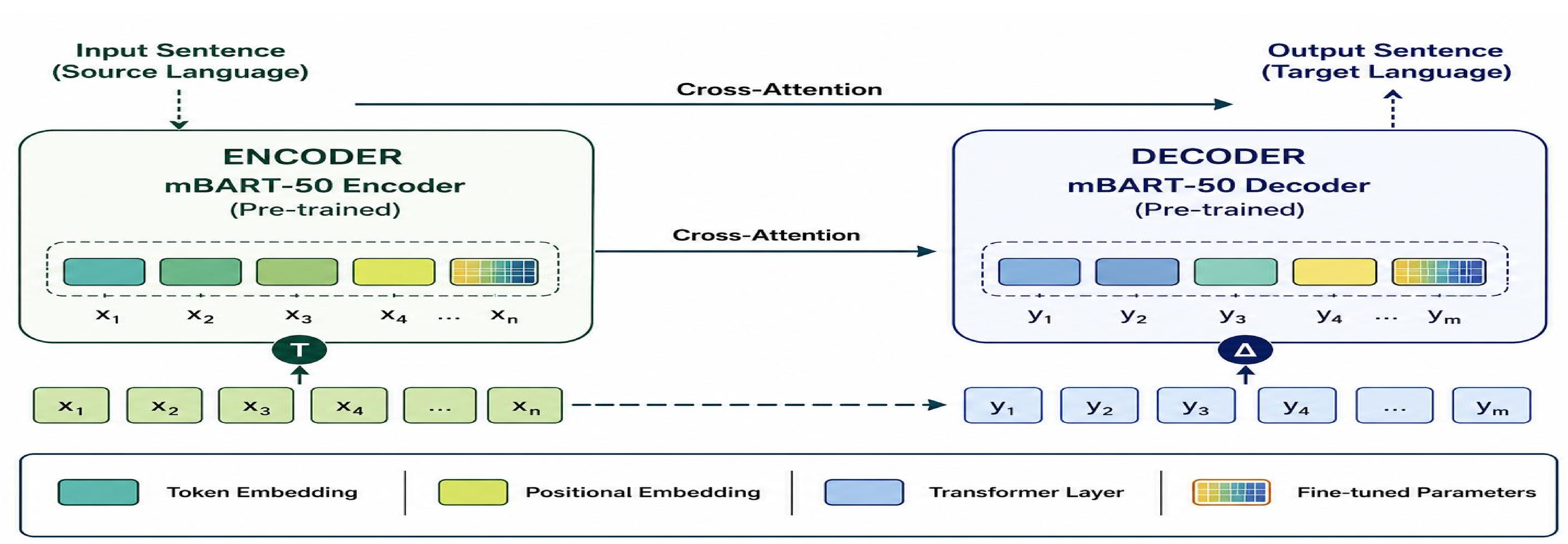
2 Preprocessing & Cleaning
Clean text, normalize, remove noise and deduplicate.

3 Lexicon & Augmentation
Build lexicon with variations and apply lexicon-based augmentation.

4 Model Fine-tuning (mBART-50)
Fine-tune mBART-50 for both translation directions.

5 Evaluation (Automatic + Human)
Evaluate using BLEU, chrF, and conduct human evaluation.

6 Deployment (Flutter + FastAPI)
Deploy model via FastAPI and build Flutter mobile app.



EXPERIMENTAL DESIGN (FOUR MODELS)

MODEL 1 — BN → CTG (Original)
mBART-50 base
Fine-tuned on original dataset
✓ mBART-50 base

MODEL 2 — CTG → BN (Original)
mBART-50 base
Fine-tuned on original dataset
✓ mBART-50 base

MODEL 3 — BN → CTG (Augmented)
mBART-50 base
Fine-tuned on auamented dataset
✓ mBART-50 base

MODEL 4 — CTG → BN (Augmented)
mBART-50 base
Fine-tuned on augmented dataset
✓ mBART-50 base

AUTOMATIC EVALUATION RESULTS

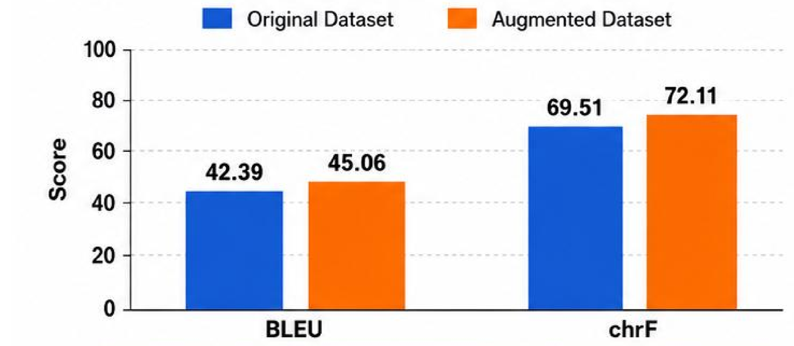
Evaluation Metrics: BLEU (higher is better) | chrF (higher is better)

1. STANDARD BANGLA → CHATGAYA

(Translation Direction: Standard Bangla to Chatgaya)

Training Dataset	BLEU	chrF
Original Dataset	42.39	69.51
Augmented Dataset	45.06	72.11

PERFORMANCE COMPARISON



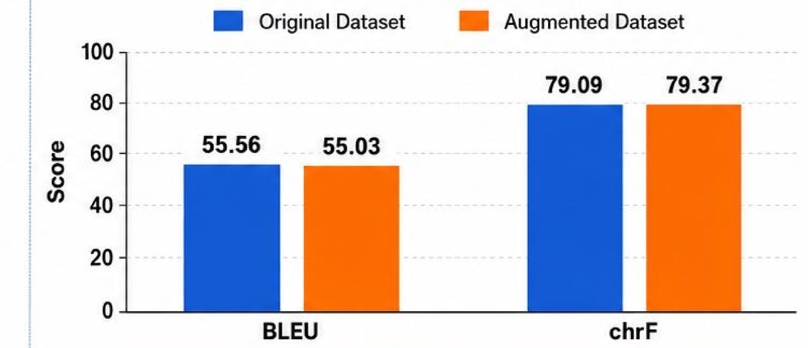
IMPROVEMENT (AUGMENTED vs ORIGINAL)
BLEU +2.67 | chrF +2.60

2. CHATGAYA → STANDARD BANGLA

(Translation Direction: Chatgaya to Standard Bangla)

Training Dataset	BLEU	chrF
Original Dataset	55.56	79.09
Augmented Dataset	55.03	79.37

PERFORMANCE COMPARISON



IMPROVEMENT (AUGMENTED vs ORIGINAL)
BLEU -0.53 | chrF +0.28

KEY TAKEAWAY

For Standard Bangla → Chatgaya, data augmentation improves both BLEU and chrF scores significantly.

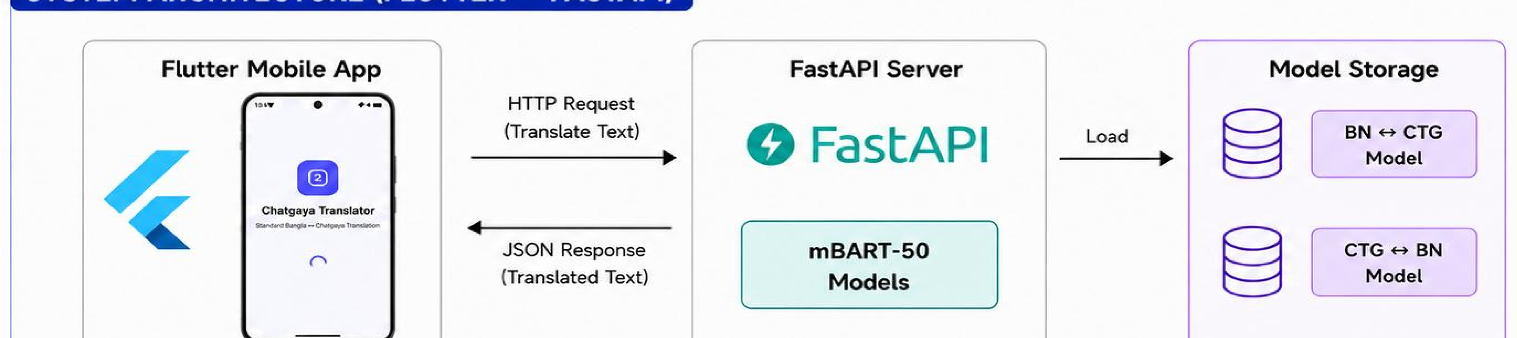
For Chatgaya → Standard Bangla, chrF shows a slight improvement while BLEU remains almost stable.

HUMAN EVALUATION (FLUENCY & ADEQUACY)

Model 1	Model 2	Model 3	Model 4
3.91 / 5	4.02 / 5	4.31 / 5	4.52 / 5

Scores are based on native speaker evaluation on fluency and adequacy (1 = Poor, 5 = Excellent)

SYSTEM ARCHITECTURE (FLUTTER ↔ FASTAPI)



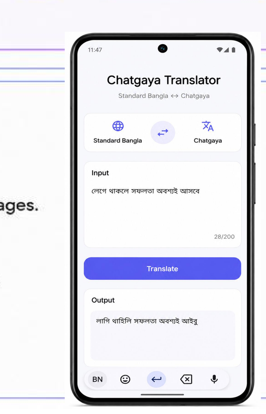
APPLICATION SCREENSHOTS

▶ Splash / Home Screen

App splash screen with application logo and title. Indicates loading state while preparing the translator.

▶ Translator Screen

Main translation interface.
- Select source and target languages.
- Enter text to translate.
- Get translated output instantly.
- Additional quick controls for better user experience.



RESEARCH CONTRIBUTIONS

- Developed a **manually curated Standard Bangla–Chatgaya parallel corpus** for low-resource NMT.
- Investigated the effectiveness of **lexicon-based data augmentation** for bidirectional translation.
- Evaluated four fine-tuned **mBART-50** models using BLEU, chrF, and human evaluation.
- Implemented a **Flutter–FastAPI** prototype for real-time bidirectional translation.

LIMITATIONS

- Parallel corpus size is relatively small compared with large-scale NMT datasets.
- Evaluation is limited to the **Chatgaya dialect** only.
- Current prototype supports **sentence-level text translation** only.
- Speech input, speech synthesis, and offline translation are not included.

FUTURE ENHANCEMENT

- Expand the parallel corpus with larger and more diverse sentence pairs.
- Investigate advanced multilingual Transformer models and efficient fine-tuning methods.
- Extend the system with **speech-to-text**, **text-to-speech**, and offline inference.
- Adapt the framework to other Bangla regional dialects and cloud deployment.

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